

MD. RAFIUL ISLAM

Computer Science and Engineering Student
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PROFILE STATEMENT

Computer Science and Engineering student with hands-on experience across web development, machine learning, and computer graphics. Track record of building full-stack applications with Flask and Django, implementing 3D rendering systems with PyOpenGL, and developing federated learning architectures for real-world classification tasks. Comfortable owning a project end to end, from system design through deployment and presentation.

SKILL SETS

- **Languages & Web:** Python, C/C++, JavaScript, HTML/CSS, Flask, Django
- **Databases & Tools:** MySQL, SQLAlchemy, Pytest, Git, PyOpenGL, GLUT
- **Machine Learning:** Federated Learning, Variational Autoencoders (VAEs), Dirichlet Partitioning, Hierarchical Models
- **Networking & OS:** Cisco Routing, VLSM, Fedora Linux, Windows

PROJECTS

Federated Learning with Hierarchical Buffer Clustering (Thesis)

July '25 – June '26

Python, PyTorch, Federated Learning

- Developed a federated learning framework across six heterogeneous clients sharing a common classifier head via FedAvg with FedProx regularization.
- Designed a hierarchical clustering-based buffer (HAC on ConvNeXt features) that improved non-buffer class accuracy by 12.57 percentage points, reaching 63.76% overall test accuracy.
- Validated privacy properties of the framework, measuring 0% property-inference attack accuracy.
- Co-authoring an IEEE conference paper and an Elsevier journal article based on this work.

AI Research Assistant | [Code](#)

FastAPI, Flask, PostgreSQL, SQLAlchemy, Gemini API

- Built a decoupled full-stack platform helping researchers manage literature, collaborate on projects, and accelerate reviews with AI, deployed live on Vercel.
- Integrated the Gemini API to auto-generate paper summaries, abstracts, and methodology breakdowns, plus a literature matrix for comparing variables across papers.
- Designed a FastAPI/Pydantic backend with PostgreSQL (Supabase) and Alembic migrations, paired with a Flask and Tailwind CSS frontend.
- Added project collaboration tools including shared workspaces, Kanban task boards, and a supervisor portal for milestone tracking and feedback.

Medicare: Online Pharmacy Platform | [Code](#)

Python, Flask, MySQL, SQLAlchemy, JavaScript

- Architected a multi-role online pharmacy platform (admin, staff, customer) with Python, Flask, and MySQL, following MVC design principles.
- Built secure authentication, role-based dashboards, and a full e-commerce flow with real-time cart management and bKash payment integration.
- Added real-time customer support chat and a Pytest-based backend test suite to ensure system reliability.

ZK Network: Secure Messaging Platform | [Code](#)

Python, Django, Cryptography, SQLite

- Developed an end-to-end encrypted messaging platform for a CSE447 Cryptography course, implementing hybrid asymmetric cryptography (RSA, ECC) and server-trusted key escrow from scratch.
- Secured data at rest with zero-trust SQLite storage, enforcing strict 2FA (Email OTP), stateless JWT sessions, and HMAC-SHA256 for message integrity against database tampering.
- Engineered a custom Django management command for a secure master key rotation protocol, actively debugging and resolving associated errors to ensure seamless, atomic data re-encryption.

EXPERIENCE

Dept. of CSE, BRAC University

Mar '26 – Present

Undergraduate Student Tutor

Dhaka, Bangladesh

- Appointed as Student Tutor — a merit-based role equivalent to Teaching Assistant — recognizing strong academic performance and dedication to peer education.
- Lead structured consultation sessions helping students grasp core course concepts and prepare effectively for assignments, quizzes, and exams.
- Collaborate closely with faculty to align tutoring efforts with course objectives and drive measurable improvements in student learning outcomes.

ACHIEVEMENTS

- Awarded a merit-based scholarship at BRAC University for academic excellence.
- Solved over **200+ problems** on Codeforces and LeetCode.

EDUCATION

BRAC University <i>Bachelor of Science in Computer Science and Engineering, CGPA: 3.93 out of 4.00</i>	Sep '26 (Expected) <i>Dhaka, Bangladesh</i>
Birshreshtha Munshi Abdur Rouf Public College <i>HSC - Higher Secondary Certificate (Science), GPA: 5.00 out of 5.00</i>	Jun '19 – Dec '21 <i>Dhaka, Bangladesh</i>
Birshreshtha Munshi Abdur Rouf Public College <i>SSC - Secondary School Certificate (Science), GPA: 4.83 out of 5.00</i>	Jan '17 – Feb '19 <i>Dhaka, Bangladesh</i>

TEAMWORK & COLLABORATION

- Collaborated within a cross-functional academic project team, coordinating shared deliverables and rebalancing workload across members to meet deadlines.
- Prepared network configuration overviews and routing methodology presentations for project vivas and academic showcases.

INTERESTS

- Pushing the boundaries of competitive programming to solve complex algorithmic challenges.
- Sharing knowledge and supporting peer learning through tutoring and hands-on teaching.
- Exploring the latest advancements in AI and applying it to real-world classification and research problems.
- Developing impactful web applications, using modern full-stack technologies to enhance user experiences.